Twin Arithmetic and Monogenic Transport of Ring Structures

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Abstract

We develop a general theory of arithmetic hierarchies obtained by transporting the algebraic structure of a ring through the action of a monogenic group of applications. Given a bijection $\phi : A \to A$ on a ring $(A, +, \cdot)$, the iterates ϕ^n induce a family of conjugate operations $(\oplus_n, \otimes_n)$ forming a hierarchy of "twin rings", all isomorphic to the base ring. This indexing by the additive group $(\mathbb{Z}, +)$ leads to a canonical theory of prime operations, factorization of operations, base-change identities, and arithmetical decompositions mirroring the classical decomposition of integers into prime factors. The exponential case $\phi = \exp$ recovers ordinary addition and multiplication as two consecutive levels of the same structure, while higher levels produce non-linear arithmetic systems. This article establishes the algebraic foundations of these twin hierarchies and prepares the study of growth laws, combinatorics, and higher-order arithmetics.

1 Introduction

In the classical approach, a binary operation

$$\odot : E \times E \to E$$

is a fixed algebraic object defined in a single coordinate system. In the framework of *Twin Spaces*, mathematical structures are no longer studied in isolation but as families indexed by a symmetry group acting on the underlying set. Given a group G acting on E, the fundamental compatibility rule for an equivariant operation reads

$$(x \odot_{e,g} y) = \overline{g}\left(x \odot_e y\right),$$

where e is the identity of G, and $\overline{g} = g^{-1}$.

In this article we focus on the case where the symmetry group is *monogenic*, generated by a bijection $\phi : E \to E$. Thus,

$$G = \langle \phi \rangle = \{\phi^n : n \in \mathbb{Z}\}$$

with composition as group law. This group is always isomorphic to $(\mathbb{Z}, +)$, and its index n acts as an "arithmetical coordinate" used to generate a hierarchy of operations and, more generally, entire algebraic structures.

Transported arithmetic. Given a reference operation $\odot_0$ on E, the twin transport rule produces, for each integer n,

$$x \odot_n y = \phi^n\big(\phi^{-n}(x) \odot_0 \phi^{-n}(y)\big).$$

The family $\{\odot_n\}_{n \in \mathbb{Z}}$ forms a hierarchy of conjugate operations. Every level is algebraically equivalent to the reference level via ϕ^{-n}, and the hierarchy inherits the additive structure of $\mathbb{Z}$.

Extension to rings. The construction extends verbatim to any ring $(A, +, \cdot)$:

$$x \oplus_n y = \phi^n\big(\phi^{-n}(x) + \phi^{-n}(y)\big), \qquad x \otimes_n y = \phi^n\big(\phi^{-n}(x) \cdot \phi^{-n}(y)\big),$$

producing twin rings $(A, \oplus_n, \otimes_n)$ all isomorphic to $(A, +, \cdot)$.

Base-change and primality. The group isomorphism $G \simeq \mathbb{Z}$ allows transportation of classical arithmetic to the level of operations:

$$x_1 \odot_{n+k} x_2 = \phi^k\big(\phi^{-k}(x_1) \odot_n \phi^{-k}(x_2)\big).$$

The integer index also carries the *multiplicative* structure of $\mathbb{N}^*$: by the iteration identity $\phi^{ab} = (\phi^a)^{\circ b}$, a factorization $q = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ realizes $\odot_q$ as a nesting of the prime-level constructions $\odot_{p_i}$. This is unique factorization in $(\mathbb{N}^*, \times)$ — a structure on the index distinct from the additive group law $\phi^n\phi^m = \phi^{n+m}$ of $G \simeq (\mathbb{Z}, +)$.

The exponential case. When $\phi = \exp$ (on any invertible domain), the twin hierarchy recovers:

$$x \odot_0 y = x + y, \qquad x \odot_1 y = xy,$$

and higher levels yield non-linear "exponential" arithmetics. Addition and multiplication appear as two consecutive states in the same transported structure.

Historical and conceptual motivation

The idea of iterating operations is as old as classical hyper-operations and the Ackermann hierarchy. However, these constructions are usually defined syntactically rather than algebraically and depend on specific choices of analytic functions.

The twin approach is different in nature:

- It applies to *any* ring, not only to $\mathbb{R}$ or to analytic settings.
- It is driven by *symmetry* rather than by combinatorial iteration.
- It produces a *family of isomorphic algebraic structures*, not a single growing function.
- It yields a true *arithmetic of operations*, including primality, factorization, and equivalence classes.

This makes twin arithmetic both a generalization of classical operations and a new framework for understanding algebraic transformations through group actions.

The goal of this article is to lay the algebraic foundations of twin arithmetic and prepare for the study of higher-level structures, including growth laws, combinatorial identities, and applications.

2 Twin Structures on Rings

2.1 Monogenic actions and transported ring operations

Let $(A, +, \cdot)$ be any ring (commutative or not). Let $\phi : A \to A$ be a bijection. We define the monogenic group

$$G = \langle \phi \rangle = \{\phi^n : n \in \mathbb{Z}\}$$

acting on A by:

$$\phi^n \cdot x = \phi^n(x).$$

Definition 2.1 (Twin ring operations)**.** For each $n \in \mathbb{Z}$ define

$$x \oplus_n y := \phi^n\big(\phi^{-n}(x) + \phi^{-n}(y)\big), \qquad x \otimes_n y := \phi^n\big(\phi^{-n}(x) \cdot \phi^{-n}(y)\big).$$

Proposition 2.2 (Twin rings)**.** *Each* $(A, \oplus_n, \otimes_n)$ *is a ring. Moreover,*

$$\Phi_n : A \to A, \quad \Phi_n(x) = \phi^{-n}(x)$$

yields an isomorphism

$$\Phi_n : (A, \oplus_n, \otimes_n) \xrightarrow{\sim} (A, +, \cdot).$$

Proof. The map $\Phi_n = \phi^{-n}$ is a bijection of A, and by definition $\Phi_n(x \oplus_n y) = \phi^{-n}\phi^n\big(\phi^{-n}(x) + \phi^{-n}(y)\big) = \Phi_n(x) + \Phi_n(y)$, and likewise $\Phi_n(x \otimes_n y) = \Phi_n(x) \cdot \Phi_n(y)$. Thus Φ_n carries $(A, \oplus_n, \otimes_n)$ bijectively onto $(A, +, \cdot)$ while preserving both operations; transporting the ring axioms back along $\Phi_n^{-1} = \phi^n$ shows $(A, \oplus_n, \otimes_n)$ is a ring, with zero $\phi^n(0)$, unit $\phi^n(1)$ and additive inverse $\ominus_n x = \phi^n(-\phi^{-n}x)$. Hence every twin level is a ring, isomorphic to the base ring via Φ_n. □

2.2 Base-change identity and factorization

Proposition 2.3 (Base-change identity)**.** *For all* $n, k \in \mathbb{Z}$,

$$x_1 \odot_{n+k} x_2 = \phi^k\big(\phi^{-k}(x_1) \odot_n \phi^{-k}(x_2)\big).$$

Proof. Unwinding the right-hand side through the defining formula,

$$\phi^k\big(\phi^{-k}(x_1) \odot_n \phi^{-k}(x_2)\big) = \phi^k\phi^n\big(\phi^{-n}\phi^{-k}(x_1) \odot_0 \phi^{-n}\phi^{-k}(x_2)\big) = \phi^{n+k}\big(\phi^{-(n+k)}(x_1) \odot_0 \phi^{-(n+k)}(x_2)\big),$$

which is $x_1 \odot_{n+k} x_2$, using $\phi^k\phi^n = \phi^{n+k}$ and $\phi^{-n}\phi^{-k} = \phi^{-(n+k)}$. □

The group law of $G \simeq (\mathbb{Z}, +)$ is additive: composition shifts the level, $\phi^n\phi^m = \phi^{n+m}$, and the base-change identity expresses this. A separate, *multiplicative* structure on the index arises from iteration, $\phi^{ab} = (\phi^a)^{\circ b}$; with respect to it the levels inherit the prime/composite behavior of $(\mathbb{N}^*, \times)$.

Definition 2.4 (Prime and composite operations)**.** For $n \geq 2$ we call $\odot_n$ *prime* if the integer n is prime — equivalently, if ϕ^n is not a non-trivial iterate $(\phi^d)^{\circ(n/d)}$ for any $1 < d < n$ — and *composite* otherwise. This refers to the multiplicative structure of the index $n \in \mathbb{N}^*$, not to the additive group law of G.

Proposition 2.5 (Iterated structure of composite levels)**.** *Let* $q = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ *be the prime factorization of* q *in* $(\mathbb{N}^*, \times)$*. By the iteration identity* $\phi^{ab} = (\phi^a)^{\circ b}$*, the operation* $\odot_q$ *is the corresponding nesting of the prime-level twin constructions: writing* $\psi = \phi^a$*, the level-*ab *operation* $\odot_{ab}$ *equals the level-*b *twin built from* ψ*. This nesting is unique up to the order of the (commuting) factors, the uniqueness being that of the integer factorization of* q*. It is* not *a decomposition under the additive group law* $\phi^n\phi^m = \phi^{n+m}$ *of* $G \simeq (\mathbb{Z}, +)$*.*

Proof. Write $\psi = \phi^a$, so that $\psi^{\circ b} = (\phi^a)^{\circ b} = \phi^{ab}$. The level-$b$ twin operation built from the bijection ψ is $x \mapsto \psi^b\big(\psi^{-b}(x) \odot_0 \psi^{-b}(y)\big) = \phi^{ab}\big(\phi^{-ab}(x) \odot_0 \phi^{-ab}(y)\big) = x \odot_{ab} y$. Applying this with the successive prime-power factors of q realizes $\odot_q$ as the iterated nesting of the prime-level constructions. The factors commute and the factorization of q is unique (fundamental theorem of arithmetic), so the nesting is unique up to the order of factors. □

Remark 2.6. The two structures on the index must not be conflated. As a *group*, $G = \langle\phi\rangle \simeq (\mathbb{Z}, +)$ is generated by ϕ and has no notion of primality; the base-change identity expresses the additive (torsor) structure of $\{\odot_n\}$. Primality and factorization live in the *multiplicative* monoid $(\mathbb{N}^*, \times)$ and are realized on operations through iteration of the transport, $\phi^{ab} = (\phi^a)^{\circ b}$.

3 Hierarchies from Monogenic Transport

Let $(A, +, \cdot)$ be a ring, and $\phi : A \to A$ a bijection. We obtain the transported operations

$$x \odot_n y = \phi^n\big(\phi^{-n}(x) \odot_0 \phi^{-n}(y)\big).$$

3.1 The exponential case

When $\phi = \exp$ on a suitable domain,

$$x \odot_0 y = x + y, \qquad x \odot_1 y = xy, \qquad x \odot_2 y = \exp(\log(x)\log(y)).$$

Higher levels produce non-linear arithmetic systems with structurally inherited properties.

Remark 3.1 (Domains in the exponential case)*.* On $\mathbb{R}$ the map exp is a bijection only onto $\mathbb{R}_{>0}$, so the iterates $\phi^{\pm n}$, and hence $\odot_n$, are defined only on the nested domains where the relevant logarithms exist; for example $\odot_2(x, y) = \exp(\log x \, \log y) = x^{\log y}$ requires $x, y > 0$. As with all iterated exp / log towers, each higher level shrinks the natural domain. On a ring where ϕ is a genuine bijection (a formal, p-adic, or finite setting) no such restriction arises and the whole hierarchy $\{\odot_n\}_{n\in\mathbb{Z}}$ is globally defined.

3.2 Aggregated operations

If $\odot_0$ is associative,

$$\bigodot_{i=1}^{k}{}^{(n)} x_i = \phi^n\left(\bigodot_{i=1}^{k}{}^{(0)} \phi^{-n}(x_i)\right),$$

generalizing exponential aggregation, log-domain linearization, and other classical transforms.

4 Growth Laws and Higher-order Operations

Proposition 4.1 (Order preservation)**.** *Suppose A is totally ordered and ϕ is strictly increasing. Then for every n the twin operation $\odot_n$ is strictly increasing in each argument, and $\phi^{-n} : (A, \odot_n) \to (A, \odot_0)$ is an order isomorphism. In particular all levels share the monotonicity type of the base operation.*

Proof. The maps ϕ^n and ϕ^{-n} are strictly increasing (composites of a strictly increasing bijection), and $x \odot_n y = \phi^n(\phi^{-n}x \odot_0 \phi^{-n}y)$ is a composite of increasing maps with the increasing $\odot_0$, so it is strictly increasing in each argument. The displayed map is an order isomorphism by the Twin-rings proposition together with this monotonicity. □

Remark 4.2 (The neutral elements move with the level)*.* The level-n ring has zero $0_n = \phi^n(0)$, unit $1_n = \phi^n(1)$, additive inverse $\ominus_n x = \phi^n(-\phi^{-n}x)$, and, where defined, multiplicative inverse $\phi^n\big((\phi^{-n}x)^{-1}\big)$. The neutral elements therefore translate with n: for $\phi = \exp$ the additive zero $0_0 = 0$ becomes the multiplicative unit $1_1 = e^0 = 1$ at the next level, which is precisely why level-0 addition becomes level-1 multiplication.

Remark 4.3 (Twin hierarchy versus hyperoperations)*.* For $\phi = \exp$ the sequence $\odot_0 = +$, $\odot_1 = \times$ continues with $\odot_2(x, y) = x^{\log y}$, a commutative *associative* third operation conjugate to $+$. This is *not* the Ackermann/Goodstein hyperoperation sequence, whose third term (tetration) is already neither associative nor commutative. Every twin level remains a ring operation isomorphic to the base; the two sequences agree only at levels 0 and 1. The "growth" of a level is thus entirely carried by the stretching of ϕ^n, not by a change in algebraic type.

5 Prime Decompositions and Twin Complexity

Definition 5.1 (Twin complexity)**.** For $q \geq 1$ define the *twin complexity* $\Omega(\odot_q) := \Omega(q)$, the number of prime factors of q counted with multiplicity, with $\Omega(\odot_1) = 0$.

Proposition 5.2 (Minimal nesting depth)**.** *The minimal number of prime-level twin constructions whose iterated nesting realizes $\odot_q$ equals $\Omega(q)$.*

Proof. By the proposition on iterated structure, each prime factor p of q contributes exactly one level-p nesting, so a factorization of q into k primes (with multiplicity) yields a nesting of depth k. The fundamental theorem of arithmetic gives $k = \Omega(q)$. For minimality, m prime constructions multiply the index by a product of m primes, hence reach an index with at most m prime factors; reaching q therefore requires $m \geq \Omega(q)$. □

Remark 5.3 (Additive cost versus twin complexity)*.* Twin complexity measures cost in the multiplicative monoid $(\mathbb{N}^*, \times)$ — the nesting depth of transports. The additive group $G \simeq (\mathbb{Z}, +)$ carries a different cost: the word length $|n|$ of ϕ^n in the generator ϕ. The two are unrelated. For instance $\odot_{2^{10}}$ has additive cost 1024 but twin complexity 10, whereas $\odot_p$ for a large prime p has additive cost p but twin complexity 1.

6 Applications

The twin hierarchy exhibits several familiar constructions as instances of a single transport.

Log-domain arithmetic and numerical stability. For $\phi = \exp$, level 1 is multiplication and level 0 is addition, so evaluating a product in the $\phi^{-1} = \log$ coordinate replaces multiplication by addition — the standard log-domain trick. The level-(-1) operation $x \odot_{-1} y = \log(e^x + e^y)$ is the *LogSumExp* function, the numerically stable evaluation of a sum of exponentials ubiquitous in statistics and machine learning. Both are twin operations of the single bijection exp.

Relativistic addition. For $\phi = \operatorname{artanh}(\,\cdot\,/c)$ on $(-c, c)$ with $\odot_0 = +$, level 1 is Einstein velocity addition $x \odot_1 y = (x + y)/(1 + xy/c^2)$, and the hierarchy iterates the rapidity map. The associativity of velocity addition is inherited, through ϕ, from associativity of $+$.

q-arithmetic. Taking ϕ a q-exponential recovers the q-addition and q-multiplication of nonextensive statistics [3] as adjacent twin levels, so that their algebraic identities are transported instances of the ordinary ring axioms.

In each case the value of the twin formalism is uniformity: a single bijection ϕ and its integer iterates organize, as one hierarchy, operations that are classically studied in isolation.

References

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[3] C. Tsallis, *Introduction to Nonextensive Statistical Mechanics*, Springer, 2009.